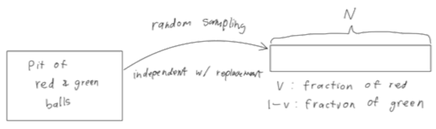
Generalization Theory

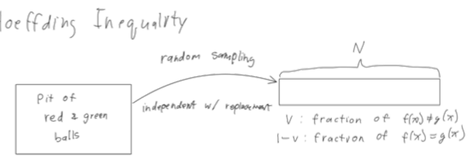
Can we make in-sample error small enough?

Can we ensure out-of-sample error is close enough to the in-sample error



predict μ, probability of retrieving a green ball $\quad P(|v-\mu| > \varepsilon) \leq 2^{(-2\varepsilon^2 N)}$

Hoeffding Inequality



predict μ, probability of retrieving $f(x) \neq g(x)$

$1-\mu$, probability of retrieving $f(x) = g(x)$

InSample Error — outOfSample Error

$$P(|E_{in}(g) - E_{out}(g)| > \varepsilon) \leq 2^{(-2\varepsilon^2 N)}$$

Limitation of Hoeffding Inequality

$$P(|E_{in}(g) - E_{out}(g)| > \varepsilon) \leq 2^{(-2\varepsilon^2 N)}$$

relies on: in-sample dataset & Hypothesis function

unknown non-random value dependent on Hypothesis function

- Hypothesis function must be selected before generating the in-sample dataset for inequality to hold.
- Paradox

notability

Solving inSample Dataset and Hypothesis Paradox

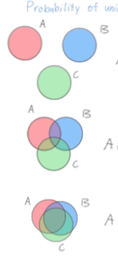
Boole's Inequality

- If $A \Rightarrow B$ then $P(A) \leq P(B)$
- $P(A \text{ or } B \text{ or } \dots F) \leq P(A) + P(B) + \dots + P(F)$

Probability of unions events $\leq$ sum of all events' probabilities

If A depends on B means A's prob $\leq$ B's prob

Apply to Hoeffding



$A \cap B \cap C = \emptyset \Rightarrow P(A \cup B \cup C) = P(A) + P(B) + P(C)$

$A \cap B \cap C \neq \emptyset \Rightarrow P(A \cup B \cup C) < P(A) + P(B) + P(C)$

$A \cap B \cap C \neq \emptyset \Rightarrow P(A \cup B \cup C) << P(A) + P(B) + P(C)$

$$P(|E_{in}(g^*) - E_{out}(g^*)| > \varepsilon) \leq P(|E_{in}(g_1) - E_{out}(g_1)| > \varepsilon \text{ OR } \dots |E_{in}(g_n) - E_{out}(g_n)| > \varepsilon)$$
$$\leq \sum_{m=1}^{M} P(|E_{in}(g_m) - E_{out}(g_m)| > \varepsilon)$$

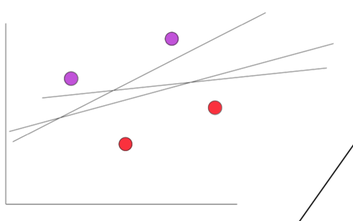
} Boole's Inequality

$$P(|E_{in}(g^*) - E_{out}(g^*)| > \varepsilon) \leq \sum_{m=1}^{M} P(|E_{in}(g_m) - E_{out}(g_m)| > \varepsilon)$$

$$P(|E_{in}(g^*) - E_{out}(g^*)| > \varepsilon) \leq \sum_{m=1}^{M} 2^{(-2\varepsilon^2 N)}$$

$$\boxed{P(|E_{in}(g^*) - E_{out}(g^*)| > \varepsilon) \leq M\left(2^{(-2\varepsilon^2 N)}\right)}$$

Linear Classification Model



Growth Function

$$m_G(N) = 2^N \leq N^{dvc} + 1$$

Vapnik Chervonenkis (VC) dimension

aka

DVC

Loose Probability Bound Vulnerability

if $M \to \infty$ aka ∞ hypothises

$$P(|E_{in}(g^*) - E_{out}(g^*)| > \varepsilon) \leq (2M)^{-2\varepsilon^2 N}$$

Solution tighter bound

$A \cap B \cap C = \phi$

$P(A \cup B \cup C) = P(A) + P(B) + P(C)$

$A \cap B \cap C \neq \phi$

$P(A \cup B \cup C) < P(A) + P(B) + P(C)$

$A \cap B \cap C \neq \phi$

$P(A \cup B \cup C) \ll P(A) + P(B) + P(C)$

GB ↓ in_samp ↑

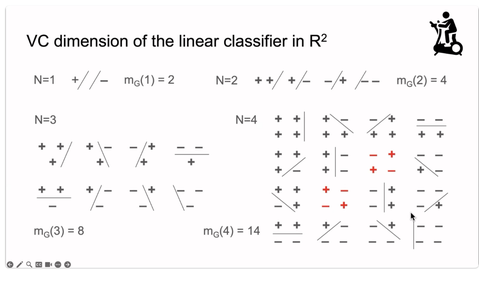
GB ↑ dvc ↑

$$E_{in}(g^*) + \sqrt{\frac{8}{N} \ln\left(\frac{4}{\delta}(2N)^{dvc} + 1\right)}$$

N (data samples), δ (tolerance), dvc (VC dimension)

$N = 1000, \quad \delta = .05, \quad dvc = 2$

$$E_{out}(g^*) \leq E_{in}(g^*) + \sqrt{\frac{8}{N} \ln\left(\frac{4}{\delta}((2N)^{dvc} + 1)\right)}$$



$g_{m,b}(x) = mx + b$

All Hypothesis functions — All effective Hypothesis functions

$G = \{g_1\ g_2 \dots \infty\}$

↓ Subset

$G_1 \{g_m \dots g_n\}$

$m_G(N)$: max # of dichotomies created by Hypothesis Set G on any N points

dvc : largest value N where $m_G(N) = 2^N$

- property of G
- measures model complexity
- $m_G(N) \leq N^{dvc} + 1$ Good Model
- $m_G(N) = 2^N$ for all $N \Rightarrow dvc(G) = \infty$ Bad Model No conclusion can be made!

$$P(E_{out}(g^*) \leq E_{in}(g^*) + GB) \geq 1 - \delta$$

GB (generalization bound)

$$GB = \sqrt{\frac{8}{N} \ln\left(\frac{4}{\delta}((2N)^{dvc} + 1)\right)}$$

$N = 1000, \quad \delta = .05, \quad dvc = 2$

$$E_{out}(g^*) \leq E_{in}(g^*) + \sqrt{\frac{8}{N} \ln\left(\frac{4}{\delta}((2N)^{dvc} + 1)\right)}$$

$dvc = 2$

$$\sqrt{\frac{8}{1000} \ln\left(\frac{4}{.05}((2 \cdot 1000)^2 + 1)\right)}$$

=

$dvc = 3$

$$\sqrt{\frac{8}{1000} \ln\left(\frac{4}{.05}((2 \cdot 1000)^3 + 1)\right)}$$

$N = ?, \quad dvc = 2, \quad GE = .01$

$$\varepsilon = \sqrt{\frac{8}{N} \ln\left(\frac{4}{\delta}((2N)^{dvc} + 1)\right)}$$

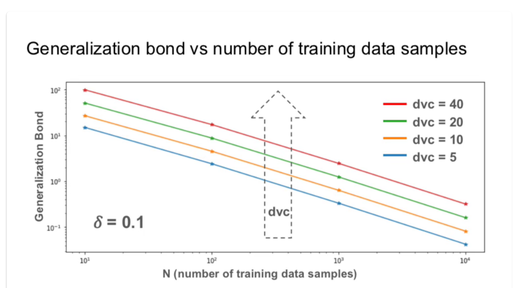
$$N = \frac{8}{\varepsilon^2} \ln\left(\frac{2N^{dvc} + 1}{\delta}\right)$$

$$N = \frac{8}{.01} \ln\left(\frac{2N^2 + 1}{.1}\right)$$

$$\varepsilon = \sqrt{\frac{8}{N} \ln\left(\frac{4}{\delta}((2N)^{dvc} + 1)\right)}$$

$$\varepsilon^2 = \frac{8}{N} \ln\left(\frac{4}{\delta}((2N)^{dvc} + 1)\right)$$

$$\varepsilon^2 N = 8 \ln\left(\frac{4}{\delta}(4(2N)^{dvc} + 1)\right)$$
$$= 8\left(\ln(4(4(2N)^{dvc} + 1) - \ln(\delta)\right)$$
$$= 8(\ln($$

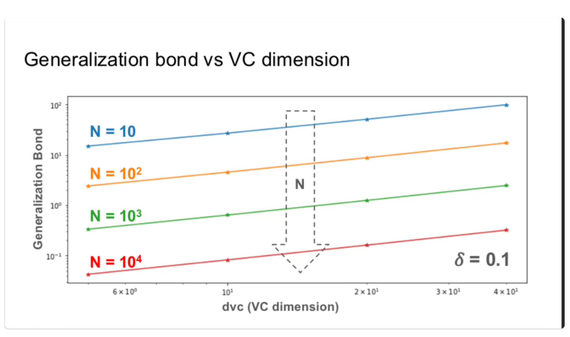


confidence = $1 - \delta$

ε = generalization error

N = amount of data samples

dvc =

$$GB = \sqrt{\frac{8}{N} \ln\left(\frac{4}{\delta}((2N)^{dvc} + 1)\right)}$$

$$N \approx \frac{8}{\varepsilon^2} \ln\left(\frac{4}{\delta}((2N)^{dvc} + 1)\right)$$